Notation used throughout

Symbol	Meaning
Ω	sample space
A, B	events
A^c	complement of A
X, Y	random variables
x_i, y_i	observed values in row i
n_A, n, N	count in A, sample size, population size
μ, σ^2, σ, p	population mean, variance, standard deviation, proportion
$\bar{X}, S^2, \widehat{P}$	estimators: sample mean, variance, proportion
$\bar{x}, s^2, \hat{p}$	estimates: their values in one sample
μ_0, p_0	values of μ and p under H_0
$\mathrm{SD}(\bar{X})$, $\mathrm{SE}(\bar{X})$	standard deviation of an estimator, and standard error: its estimate from the sample
$\mathrm{SD}_0(\widehat{P})$	standard deviation of $\widehat{P}$ under H_0
z_q, $t_{q,df}$	q quantile of $N(0,1)$ and of t_{df}: $P(Z \le z_q) = q$
z_{obs}, t_{obs}	observed value of a test statistic
α	significance level
df	degrees of freedom

Lecture 1 - Introduction to data analysis

Proportions and percentages

$$\text{proportion} = \frac{n_A}{n}, \quad \text{percentage} = 100\frac{n_A}{n}\%.$$

Sample mean and range

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \text{range} = \max_i x_i - \min_i x_i.$$

Median, quartiles, and interquartile range

Q_1, median, Q_3: values with at least 25, 50, and 75 percent of the observations at or below them, and at least 75, 50, and 25 percent at or above them.

$$\mathrm{IQR} = Q_3 - Q_1.$$

Lecture 2 - Probability foundations and random variables

Probability bounds, sample space, and complement

$$0 \le P(A) \le 1, \quad P(\Omega) = 1, \quad P(A^c) = 1 - P(A).$$

Addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Addition rule for disjoint events

$$A \cap B = \varnothing \implies P(A \cup B) = P(A) + P(B).$$

Equally likely outcomes

$$P(A) = \frac{|A|}{|\Omega|} \quad \text{(equally likely outcomes)}.$$

Three more properties

$$P(\varnothing) = 0, \quad A \subseteq B \implies P(A) \le P(B),$$
$$P(A) = P(A \cap B) + P(A \cap B^c).$$

Partition of the sample space

$$B_i \cap B_j = \varnothing \ (i \ne j), \quad \bigcup_{i=1}^{k} B_i = \Omega.$$

Lecture 3 - Discrete distributions, expectation, and variance

Probability mass function (PMF)

$$p_X(x) = P(X = x), \quad p_X(x) \ge 0, \quad \sum_{x \in \mathcal{X}} p_X(x) = 1.$$

Cumulative distribution function (CDF)

$$F_X(a) = P(X \le a) = \sum_{x \le a} p_X(x).$$

Interval probability from a CDF

$$P(a < X \le b) = F_X(b) - F_X(a).$$

Expectation and expectation of a transformation

$$E[X] = \sum_{x \in \mathcal{X}} x\, p_X(x), \quad E[g(X)] = \sum_{x \in \mathcal{X}} g(x) p_X(x).$$

Variance and standard deviation of a random variable

$$\mu = E[X], \quad \sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2] = E[X^2] - (E[X])^2,$$
$$\sigma = \mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

Shifting and rescaling a variable

$$E[aX + b] = aE[X] + b, \quad \mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X),$$
$$\mathrm{SD}(aX + b) = |a|\,\mathrm{SD}(X).$$

Bernoulli distribution

$$X \sim \text{Bernoulli}(p): \quad P(X = 1) = p, \quad P(X = 0) = 1 - p,$$
$$E[X] = p, \quad \mathrm{Var}(X) = p(1-p).$$

Sample variance and standard deviation

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \quad s = \sqrt{s^2}.$$

Lecture 4 - Conditioning, Bayes' rule, and independence

Conditional probability and multiplication rule

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)},$$
$$P(A \cap B) = P(A \mid B)P(B) \quad \text{for } P(B) > 0.$$

Law of total probability

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)P(B_i).$$

Bayes' rule

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{k} P(A \mid B_j)P(B_j)}.$$

Bayes' rule for one event B

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}, \quad P(A^c \mid B) = 1 - P(A \mid B).$$

Independence of events

$$A \text{ and } B \text{ independent} \iff P(A \cap B) = P(A)P(B).$$

Lecture 5 - Joint random variables, covariance, and correlation

Joint PMF

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

Marginal and conditional PMFs

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Conditional mean

$$E[X \mid Y = y] = \sum_x x\, p_{X|Y}(x \mid y).$$

Independence of random variables: X and Y are independent if and only if

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } x, y.$$

Population covariance and correlation

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y].$$
$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{SD}(X)\,\mathrm{SD}(Y)}, \quad -1 \le \rho_{XY} \le 1.$$

Sample covariance and correlation

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}), \quad r_{XY} = \frac{s_{XY}}{s_X s_Y}.$$

Sums of two variables

$$E[X + Y] = E[X] + E[Y],$$
$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y).$$

Sums of two independent variables

$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$

Lecture 6 - Continuous distributions and the normal model

Probabilities for a continuous distribution

$f_X(x) \geq 0, \quad$ total area under $f_X = 1, \quad P(X = x) = 0,$
$P(a < X \leq b) = F_X(b) - F_X(a).$

Uniform distribution

$$X \sim U(a,b): \quad f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise}, \end{cases}$$

$$P(c < X < d) = \frac{d-c}{b-a}, \quad E[X] = \frac{a+b}{2},$$

$$\mathrm{Var}(X) = \frac{(b-a)^2}{12}, \quad \mathrm{SD}(X) = \frac{b-a}{\sqrt{12}}.$$

Normal distribution and standardisation

$X \sim N(\mu, \sigma^2), \quad Z = \dfrac{X-\mu}{\sigma} \sim N(0,1),$
$P(Z > z) = 1 - P(Z \leq z), \quad P(Z \leq -z) = P(Z > z),$
$x = \mu + z\sigma.$

Quantiles and the central part of the standard normal distribution

$P(X \leq x_q) = q, \quad x_q = \mu + z_q\sigma,$
$P\big(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2}\big) = 1-\alpha.$

Lecture 7 - Samples, parameters, and estimators

Sample mean and sample proportion estimators

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad \widehat{P} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad (X_i \in \{0,1\}).$$

Sample variance and degrees of freedom

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \quad \sum_{i=1}^{n}(X_i - \bar{X}) = 0, \quad df = n-1.$$

Shortcut for the sample variance, and the variance of a 0/1 variable

$$S^2 = \frac{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2}{n-1},$$

$X_i \in \{0,1\}: \quad \sigma^2 = p(1-p),$ estimate $\hat{p}(1-\hat{p})$.

$X_i \in \{0,1\}: \quad s^2 = \dfrac{n}{n-1}\,\hat{p}(1-\hat{p}).$

This s^2 is the value that `var()` gives for the 0/1 values.

Unbiased estimators

$E[\bar{X}] = \mu, \quad E[\widehat{P}] = p, \quad E[S^2] = \sigma^2.$

Random sample, and the 10% condition for sampling without replacement

$E[X_i] = \mu, \quad \mathrm{Var}(X_i) = \sigma^2,$
$n \leq 0.10N \implies X_1, \ldots, X_n$ treated as iid.

Lecture 8 - Sampling distributions, the central limit theorem, and the t distribution

Sample mean: standard deviation and standard error

$$E[\bar{X}] = \mu, \quad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \quad \mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}, \quad \mathrm{SE}(\bar{X}) = \frac{S}{\sqrt{n}}.$$

Distribution of the sample mean (exact for a normal population, approximate for large n)

$$\bar{X} \mathrel{\dot\sim} N\left(\mu, \frac{\sigma^2}{n}\right), \quad \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \mathrel{\dot\sim} N(0,1).$$

Sample proportion: standard deviation and standard error

$$E[\widehat{P}] = p, \quad \mathrm{SD}(\widehat{P}) = \sqrt{\frac{p(1-p)}{n}}, \quad \mathrm{SE}(\widehat{P}) = \sqrt{\frac{\widehat{P}(1-\widehat{P})}{n}},$$

value in one sample: $\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$.

Distribution of the sample proportion (approximate, if the success-failure condition holds)

$$\widehat{P} \mathrel{\dot\sim} N\left(p, \frac{p(1-p)}{n}\right), \quad Z = \frac{\widehat{P}-p}{\sqrt{p(1-p)/n}} \mathrel{\dot\sim} N(0,1).$$

Number of successes in n independent trials (binomial)

$Y \sim \mathrm{Bin}(n,p): \quad E[Y] = np, \quad \mathrm{Var}(Y) = np(1-p),$
$\widehat{P} = \dfrac{Y}{n}.$

Success-failure condition

$np \geq 10, \quad n(1-p) \geq 10.$

One-sample t statistic (normal population, or approximately for large n)

$$T = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Lecture 9 - Confidence intervals

Confidence interval for a mean and for a proportion (level $1-\alpha$)

$$\bar{X} \pm t_{1-\alpha/2,\,n-1}\frac{S}{\sqrt{n}},$$

$$\widehat{P} \pm z_{1-\alpha/2}\sqrt{\frac{\widehat{P}(1-\widehat{P})}{n}} \quad (n\hat{p} \geq 10,\ n(1-\hat{p}) \geq 10).$$

Critical values $z_{1-\alpha/2}$, and $t_{1-\alpha/2,\,df}$ for $df = 5$ to 120

$1-\alpha$	$z_{1-\alpha/2}$	df=5	10	20	30	60	100	120
0.90	1.645	2.015	1.812	1.725	1.697	1.671	1.660	1.658
0.95	1.960	2.571	2.228	2.086	2.042	2.000	1.984	1.980
0.99	2.576	4.032	3.169	2.845	2.750	2.660	2.626	2.617

If your df is not listed, use the closest listed value.

Sample size for a margin of error m around a mean (m = half the width of the interval, round up)

$$n \geq \left(\frac{z_{1-\alpha/2}\,\sigma}{m}\right)^2.$$

Confidence interval in five parts

1 formula, 2 given values, 3 calculations (standard error, margin of error), 4 interval, 5 conclusion.

Lecture 10 - Testing decisions, errors, and power

Test in five steps

1 hypotheses and α, 2 test statistic and its distribution under H_0, 3 critical value and rejection region, 4 observed value, 5 conclusion.

Two-sided test for a mean (normal population, or approximately for large n)

$H_0: \mu = \mu_0$ versus $H_1: \mu \neq \mu_0,$

$T = \dfrac{\bar{X}-\mu_0}{S/\sqrt{n}} \sim t_{n-1}$ under $H_0,$

reject H_0 if $|t_{obs}| \geq t_{1-\alpha/2,\,n-1}$.

Two-sided test for a proportion ($np_0 \geq 10$, $n(1-p_0) \geq 10$)

$H_0: p = p_0$ versus $H_1: p \neq p_0,$

$$Z = \frac{\widehat{P}-p_0}{\mathrm{SD}_0(\widehat{P})} \mathrel{\dot\sim} N(0,1) \text{ under } H_0, \quad \mathrm{SD}_0(\widehat{P}) = \sqrt{\frac{p_0(1-p_0)}{n}},$$

reject H_0 if $|z_{obs}| \geq z_{1-\alpha/2}$.

Two-sided p-value

$p = 2\,P(T \geq |t_{obs}|)$ or $2\,P(Z \geq |z_{obs}|),$
$p \leq \alpha \iff$ reject H_0.

Test and $(1-\alpha)$ confidence interval for a mean

reject $H_0: \mu = \mu_0$ at level $\alpha \iff \mu_0$ outside the interval.

Type I error, Type II error, and power

$P(\text{Type I error}) \leq \alpha, \quad \text{Power} = 1 - P(\text{Type II error}).$